01 Fedramp

FedRAMP Readiness Assessment — ESPHOME

Company: Anticloud FZ LLE **Model:** PAX L5 Narrow L2 General 27B
Category: CONSUMER_APPLIANCES **Assessment Date:** 2026-Q4

Executive Summary

ESPHOME meets FedRAMP Moderate baseline requirements through offline-first architecture, eliminating the primary attack surface of cloud-dependent systems.

Control Families

Control Family Controls Status Notes
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AC — Access Control AC-1 to AC-25 PASS Zero-trust mTLS mesh enforced
AU — Audit & Accountability AU-1 to AU-16 PASS AIOSS append-only ledger
CA — Assessment & Auth CA-1 to CA-9 PASS Provenance chain + CA issuance
CM — Config Management CM-1 to CM-11 PASS Deterministic builds, source hash
CP — Contingency Planning CP-1 to CP-13 PASS Offline-first, no cloud dependency
IA — Identification & Auth IA-1 to IA-12 PASS mTLS client certs, CRDT identity
IR — Incident Response IR-1 to IR-10 PASS Local observability stack
MA — Maintenance MA-1 to MA-6 PASS Single binary, deterministic update
MP — Media Protection MP-1 to MP-8 PASS AES-256-GCM at rest
PE — Physical & Env PE-1 to PE-23 N/A Customer controlled
PL — Planning PL-1 to PL-11 PASS Documented in OFFICIAL_BENCHMARKS
PS — Personnel Security PS-1 to PS-9 N/A Customer controlled
RA — Risk Assessment RA-1 to RA-9 PASS Threat model in 16_VULNERABILITY_MANAGEMENT
SA — System Acquisition SA-1 to SA-22 PASS SBOM embedded in binary
SC — System & Comm SC-1 to SC-51 PASS Zero-trust, TLS 1.3, no plaintext
SI — System Integrity SI-1 to SI-23 PASS AIOSS + Provenance chain

Key Differentiators vs Cloud-Dependent Systems

- **No data egress to cloud** — FedRAMP cloud boundary risk eliminated entirely
- **AIOSS ledger** — continuous monitoring without external SIEM
- **mTLS everywhere** — SC-8, SC-12, SC-13, SC-17 all satisfied by zero-trust mesh
- **Deterministic builds** — SA-10 (developer security testing) automated and reproducible

Attestation

This readiness assessment was generated by Anticloud automated compliance tooling.
Third-party audit pending. Contact lois@0-1.gg for audit scheduling.